AGRIVIVE / BAMBOO-WICK

TRIPURA / BAMBOO-CRAFT INSPIRED / TEXTILE-WICK RESERVOIR

SERIES 25 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



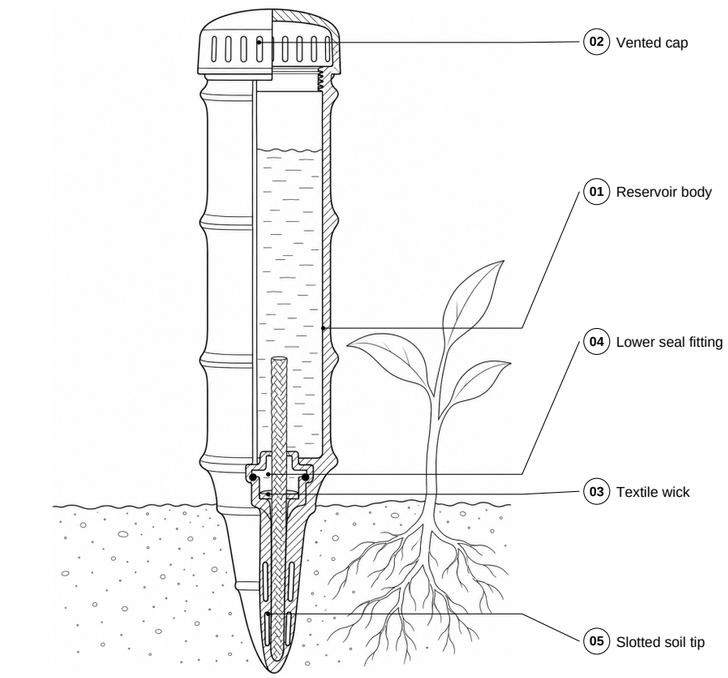
STATE CONTEXT / INDIGENOUS INSPIRATION

Bamboo and cane crafts are part of Tripura's material culture. BAMBOO-WICK uses a node-like vessel form for a new textile-wick concept; capillary irrigation is not claimed as a local indigenous practice.

Practice source: IGNCA: Bamboo and Cane Culture of Tripura | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 50 OD x 200 overall H

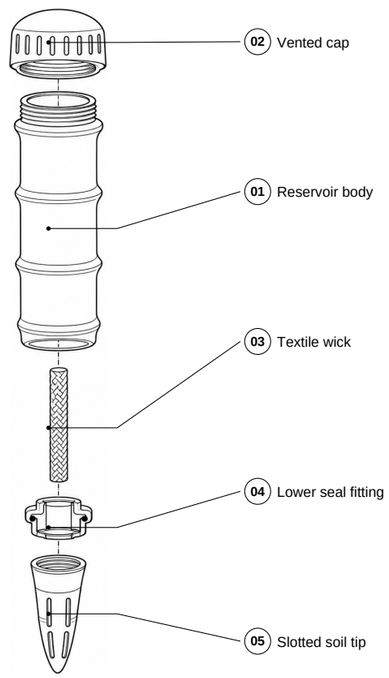


DESIGN / OPERATION NOTES

Thread the replaceable wick through the lower fitting.
Set the slotted tip in soil near a trial root zone.
Fill through the open neck, replace the vented cap and inspect the wick.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
50 OD x 200 overall H

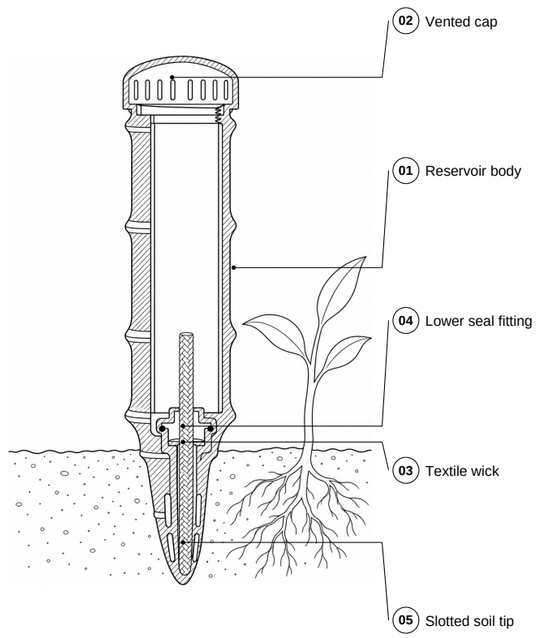


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

SECTION BASIS / mm
42 cavity ID x 150 clear H; 40 soil-tip extension; nominal vessel height 160



DESIGN / OPERATION NOTES

Set the slotted tip in soil near a trial root zone.
Fill through the open neck, replace the vented cap and inspect the wick.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Vessel OD / overall H	**50 / 200**
Cavity ID / clear H	**42 / 150**
Soil-tip extension	**40**

02 GEOMETRIC CHECK

GROSS REFILL CAVITY

207.8 mL

pi/4 x 42^2 x 150 / 1000
Subtract wick/fittings; wick delivery rate unverified

03 PARTS / QTY

01 Reservoir body	**1**
02 Vented cap	**1**
03 Textile wick	**1**
04 Lower seal fitting	**1**
05 Slotted soil tip	**1**

Body/cap/tip: candidate polymer. Wick: textile grade TBD. Lower fitting seal TBD.

04 INTERFACES / CHECKS

Test wick wetting, delivery and microbial fouling.

Check fitting seal and refill vent.

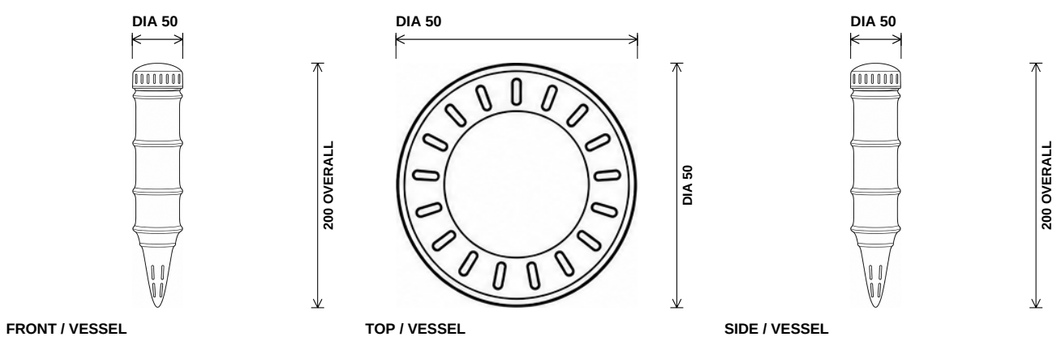
Measure working volume with wick and fittings installed.

05 DATUMS / DRAWING BASIS

A: vessel axis. B: lower seal shoulder. Check wick passage through B.

PRELIMINARY / VALIDATION PENDING

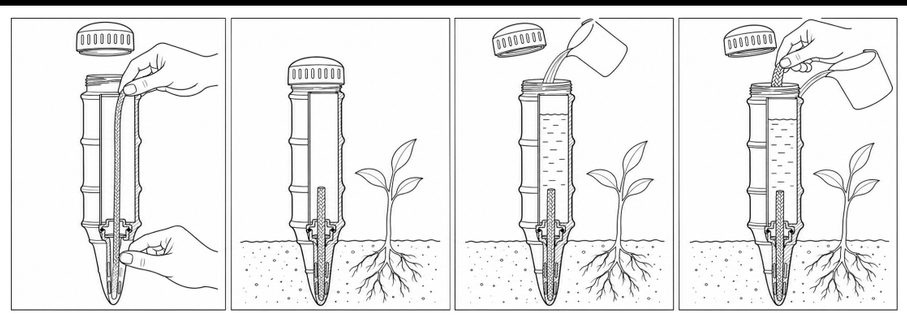
03 ORTHOGRAPHIC STUDIES / mm



FRONT / VESSEL **TOP / VESSEL** **SIDE / VESSEL**

Overall DIA 50 x 200 H: nominal vessel 160 H + tip 40. Cavity DIA 42 x 150 clear H allows end material.

05 INSTALLATION / OPERATING SEQUENCE



1 THREAD WICK **2 SET SOIL TIP** **3 OPEN / FILL** **4 INSPECT / REFILL**

Concept sequence; validate fit, access and operating conditions in prototypes.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-25 / P01	BAMBOO-WICK	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	25 / 28